

3.5 Derivative of Trigonometric Functions

Motivation:

Suppose we want to find the first derivative of $f(x) = \sin(x)$. If we use the definition of the derivative, we obtain:

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{\sin(x+h) - \sin(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x)\cos(h) + \sin(h)\cos(x) - \sin(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\sin(x)(\cos(h)-1) + \sin(h)\cos(x)}{h} = \lim_{h \rightarrow 0} \frac{\sin(x)(\cos(h)-1)}{h} + \lim_{h \rightarrow 0} \frac{\sin(h)\cos(x)}{h} \\ &= \sin(x) \lim_{h \rightarrow 0} \frac{\cos(h)-1}{h} + \cos(x) \lim_{h \rightarrow 0} \frac{\sin(h)}{h}. \end{aligned}$$

So, finding the derivative of sine comes down to computing two difficult limits. Specifically, we must compute: $\lim_{h \rightarrow 0} \frac{\cos(h)-1}{h}$ and $\lim_{h \rightarrow 0} \frac{\sin(h)}{h}$.

Theorem: If x is measured in radians, we have:

$$(1) \lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1$$

$$(2) \lim_{x \rightarrow 0} \frac{\cos(x)-1}{x} = 0$$

Proof (of 1):

We note that since x is approaching zero we may assume that x is the measure of an acute angle. Using the figure to the right, we compute 3 different areas:

$$A_{\triangle COA} = \frac{1}{2}(1)\sin(x) = \frac{1}{2}\sin(x)$$

$$A_{\text{sector}} = \frac{x}{2\pi} \cdot \pi(1)^2 = \frac{x}{2}$$

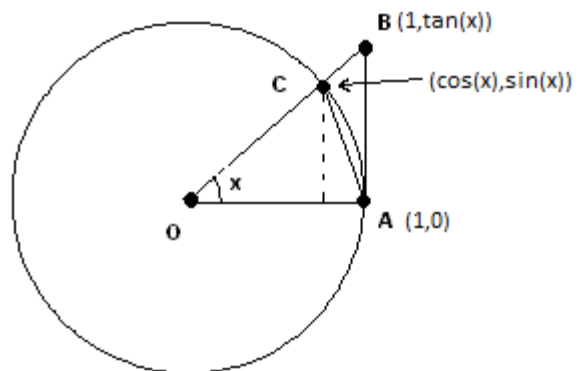
$$A_{\triangle BOA} = \frac{1}{2}(1)\tan(x) = \frac{1}{2}\tan(x)$$

From the geometry it is clear that: $A_{\triangle COA} \leq A_{\text{sector}} \leq A_{\triangle BOA}$

which implies $\frac{1}{2}\sin(x) \leq \frac{x}{2} \leq \frac{1}{2}\tan(x)$

Since $\sin(x) > 0$ when x is positive and near 0, we may multiply through by $\frac{2}{\sin(x)}$ and obtain:

$$1 \leq \frac{x}{\sin(x)} \leq \frac{1}{\cos(x)}. \text{ Taking reciprocals yields: } \cos(x) \leq \frac{\sin(x)}{x} \leq 1.$$



We also see that $\lim_{x \rightarrow 0^+} \cos(x) = 1$ & $\lim_{x \rightarrow 0^+} 1 = 1$. Therefore, by the Squeeze Theorem, we have that

$$\lim_{x \rightarrow 0^+} \frac{\sin(x)}{x} = 1. \text{ A similar argument shows that } \lim_{x \rightarrow 0^-} \frac{\sin(x)}{x} = 1 \text{ from which we may conclude that}$$

$$\lim_{x \rightarrow 0} \frac{\sin(x)}{x} = 1.$$

Proof (of 2):

We rewrite the given limit as follows:

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x} &= \lim_{x \rightarrow 0} \frac{\cos(x) - 1}{x} \cdot \frac{\cos(x) + 1}{\cos(x) + 1} = \lim_{x \rightarrow 0} \frac{\cos^2(x) - 1}{x[\cos(x) + 1]} = \lim_{x \rightarrow 0} \frac{-\sin^2(x)}{x[\cos(x) + 1]} = \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \cdot \frac{-\sin(x)}{\cos(x) + 1} \\ &= \lim_{x \rightarrow 0} \frac{\sin(x)}{x} \cdot \lim_{x \rightarrow 0} \frac{-\sin(x)}{\cos(x) + 1} = 1 \cdot \lim_{x \rightarrow 0} \frac{-\sin(x)}{\cos(x) + 1} = \frac{-\sin(0)}{\cos(0) + 1} = \frac{0}{2} = 0 \end{aligned}$$

Theorem: $\frac{d}{dx} [\sin(x)] = \cos(x)$

Proof:

Using the computation from the motivation we have that:

$$\frac{d}{dx} (\sin(x)) = \sin(x) \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} + \cos(x) \lim_{h \rightarrow 0} \frac{\sin(h)}{h} = \sin(x) \cdot 0 + \cos(x) \cdot 1 = \cos(x)$$

Note: It is CRUCIAL that x is measured IN RADIANS. The limits involved in the proof above are only valid when x is in radians. So, the derivative of sine would not be cosine if x were measured in degrees. Due to this, we always require that angles are measured in radians in calculus.

Example 1: Use the definition of the derivative to find $\frac{d}{dx} [\cos(x)]$.

(Hint: $\cos(A + B) = \cos(A)\cos(B) - \sin(A)\sin(B)$)

Using what we have shown so far we calculate:

$$\begin{aligned} \frac{d}{dx} [\cos(x)] &= \lim_{h \rightarrow 0} \frac{\cos(x+h) - \cos(x)}{h} = \lim_{h \rightarrow 0} \frac{\cos(x)\cos(h) - \sin(x)\sin(h) - \cos(x)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\cos(x)(\cos(h) - 1) - \sin(x)\sin(h)}{h} = \lim_{h \rightarrow 0} \frac{\cos(x)(\cos(h) - 1)}{h} - \lim_{h \rightarrow 0} \frac{\sin(x)\sin(h)}{h} \\ &= \cos(x) \lim_{h \rightarrow 0} \frac{\cos(h) - 1}{h} - \sin(x) \lim_{h \rightarrow 0} \frac{\sin(h)}{h} = \cos(x) \cdot 0 - \sin(x) \cdot 1 = -\sin(x) \end{aligned}$$

Example 2: Use the quotient rule to find the first derivative of each of the following.

a) $y = \sec(x)$

We note that:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{\cos(x)} \right) = \frac{\cos(x) \cdot 0 - 1 \cdot (-\sin(x))}{\cos^2(x)} = \frac{\sin(x)}{\cos^2(x)} = \frac{1}{\cos(x)} \cdot \frac{\sin(x)}{\cos(x)} = \sec(x) \tan(x).$$

b) $y = \csc(x)$

We note that:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{1}{\sin(x)} \right) = \frac{\sin(x) \cdot 0 - 1 \cdot (\cos(x))}{\sin^2(x)} = \frac{-\cos(x)}{\sin^2(x)} = \frac{-1}{\sin(x)} \cdot \frac{\cos(x)}{\sin(x)} = -\csc(x) \cot(x).$$

c) $y = \cot(x)$

We note that:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\cos(x)}{\sin(x)} \right) = \frac{\sin(x) \cdot (-\sin(x)) - \cos(x) \cdot (\cos(x))}{\sin^2(x)} = \frac{-\sin^2(x) - \cos^2(x)}{\sin^2(x)} = \frac{-1}{\sin^2(x)} = -\csc^2(x).$$

d) $y = \tan(x)$

We note that:

$$\frac{dy}{dx} = \frac{d}{dx} \left(\frac{\sin(x)}{\cos(x)} \right) = \frac{\cos(x) \cdot \cos(x) - \sin(x) \cdot (-\sin(x))}{\cos^2(x)} = \frac{1}{\cos^2(x)} = \sec^2(x).$$

Summary of Trig Function Derivatives:

$$\frac{d}{dx}[\sin(x)] = \cos(x) \quad \longleftrightarrow \quad \frac{d}{dx}[\cos(x)] = -\sin(x)$$

$$\frac{d}{dx}[\tan(x)] = \sec^2(x) \quad \longleftrightarrow \quad \frac{d}{dx}[\cot(x)] = -\csc^2(x)$$

$$\frac{d}{dx}[\sec(x)] = \sec(x)\tan(x) \quad \longleftrightarrow \quad \frac{d}{dx}[\csc(x)] = -\csc(x)\cot(x)$$

Example 3: Suppose that the position of an object on a vertical spring at time t is given by the function: $f(t) = 4\cos(t)$. Find a function for the velocity of the object at time t .

We find the velocity by taking the first derivative of the position function. So, the answer is:

$$v(t) = f'(t) = -4\sin(t)$$

Example 4: Differentiate $f(x) = x^2 \sin(x) \cos(x)$.

Notice that we have a product of three functions, so we will use the *Product Rule*. We calculate:

$$\begin{aligned} f'(x) &= \frac{d}{dx}[x^2] \sin(x) \cos(x) + x^2 \frac{d}{dx}[\sin(x)] \cos(x) + x^2 \sin(x) \frac{d}{dx}[\cos(x)] \\ &= 2x \sin(x) \cos(x) + x^2 \cos^2(x) - x^2 \sin^2(x) \\ &= x[2 \sin(x) \cos(x) + x \cos^2(x) - x \sin^2(x)] \\ &= x[\sin(2x) + x[\cos^2(x) - \sin^2(x)]] \\ &= x[\sin(2x) + x \cos(2x)] \end{aligned}$$

Note: Here we made use of two different trig. identities, namely $\sin(2x) = 2\sin(x)\cos(x)$ and $\cos(2x) = \cos^2(x) - \sin^2(x)$. It is not required that you know all such identities, but be aware that the textbook and other resources may utilize identities to simplify.